SlotSync

System Architecture Document

Clash-Free College Timetable Generator
MERN stack | Constraint-satisfaction scheduling engine | Multi-provider AI assistant

Version 1.0
Status: Approved for implementation
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At a glance

Frontend	React 18 + Vite 5 + React Router 6 + TanStack Query 5 + Zustand 4 + Framer Motion 11
Backend	Node.js + Express 4 REST API with Zod validation and JWT authentication
Scheduler	Pure-JS backtracking with forward-checking + hill-climbing optimiser
Data layer	node:sqlite / MySQL (dev) and PostgreSQL + Prisma (prod), anti-clash UNIQUE keys
AI assistant	Local VM -> OpenRouter -> Google -> NVIDIA chain with on-device fallback
Deployment	API on Render, SPA on Vercel, LLM endpoints external/self-hosted

Generated from the SlotSync repository (README + PRD).

1. Overview

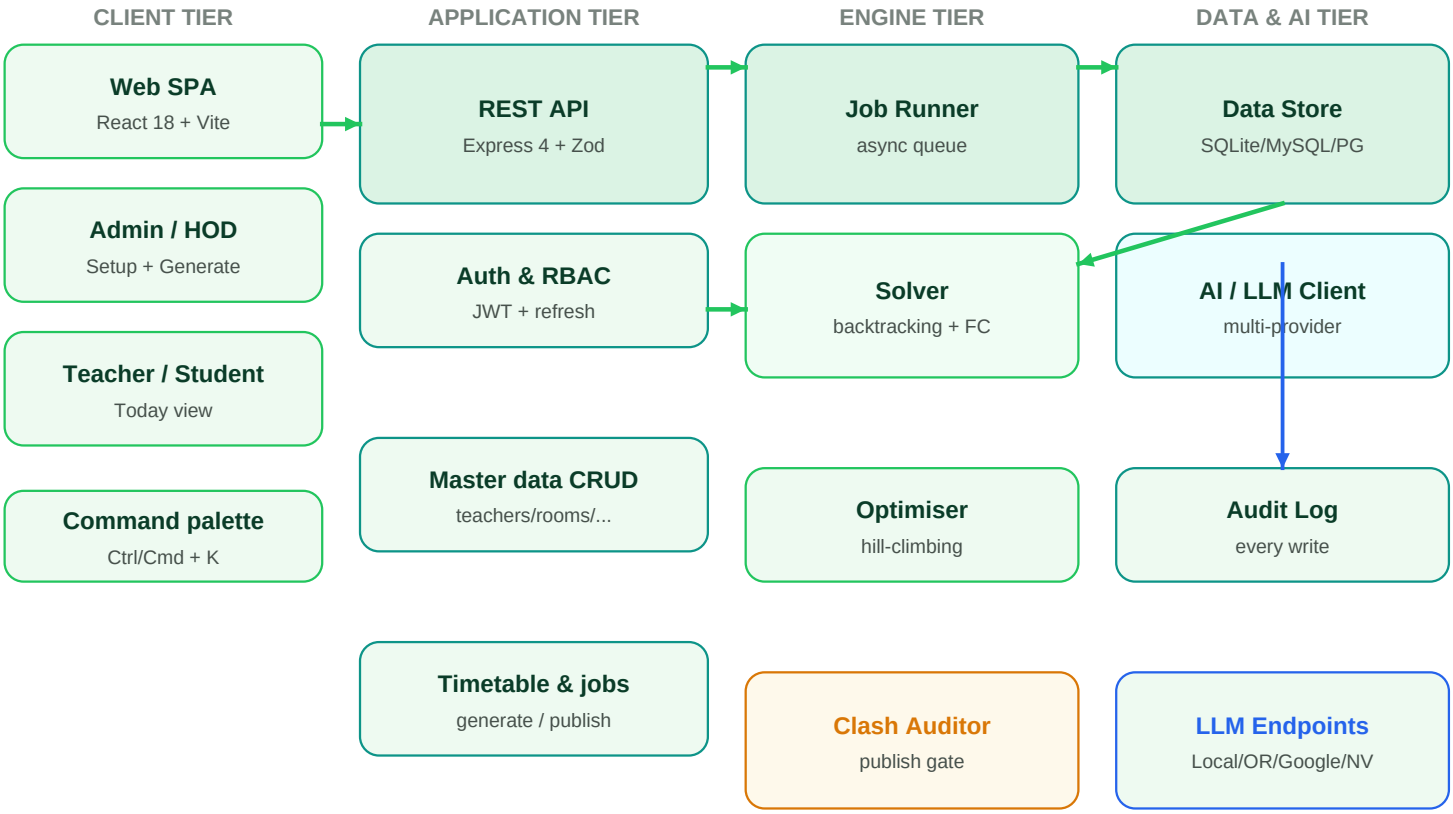
SlotSync replaces hand-built spreadsheet timetables with a one-click generator that produces a full weekly timetable with zero hard conflicts, or a plain-language report explaining exactly what blocks a solution. Administrators build the reference data once, generate a timetable, then review, edit, publish and share it. Teachers and students get a clean, mobile-first view of the same published data.

1.1 Architectural principles

- **Correctness first.** Hard constraints (no double-booked teacher, room or batch) are enforced by the solver and re-enforced by database-level UNIQUE constraints, so a clash is impossible even if a code path has a bug.
- **Separate frontend and backend.** A stateless Express REST API and a React SPA are deployed independently; the API is the only holder of business rules and credentials.
- **Async by design.** Long-running generation is offloaded to background jobs with progress polling, so the API never blocks.
- **Server-enforced authorisation.** Role and department scoping are checked on every request; hiding a UI control is never the only protection.
- **AI as an assistant, never the scheduler.** The solver and database own correctness; AI only reads, translates, explains and summarises, and never writes timetable data.
- **One source of truth per role.** Every role reads the same teachers, rooms and availability lists, so no view disagrees across roles.

2. High-Level Architecture

The system is a classic three-tier web application with a dedicated scheduling engine and an optional multi-provider AI layer. Clients talk only to the REST API; the API owns the data store, the job runner and the solver.



Reads are unified: every authenticated role receives the same full teacher, room and availability lists. Writes stay role-scoped (Admin globally, HOD within their department). Published timetables are exposed anonymously through read-only endpoints; drafts stay private.

3. Layered Architecture

Presentation Layer

React 18 SPA built with Vite. React Router for routes, TanStack Query for server state and caching, Zustand for auth/theme/view state, Framer Motion for restrained animation, dnd-kit-style grid interaction. Mobile-first; graceful loading, empty and error states; global error boundary.

API / Application Layer

Express 4 REST API. JWT authentication (15-min access token + 7-day rotating refresh cookie), Zod validation middleware, role and department guards, CORS, structured logging and a central error handler returning { error, fix } payloads.

Domain / Service Layer

Route handlers orchestrate domain services: auth, users, availability, master data, time grid, timetables, jobs, exports, audit and AI. All writes are transactional and audited.

Engine Layer

The scheduling engine expands assignments into sessions, orders them by difficulty, runs backtracking search with forward-checking, optimises soft constraints with hill-climbing, and flattens the result to one row per period. A job runner reports progress asynchronously.

Data Layer

A generic synchronous record store backed by node:sqlite or MySQL for local development, and a PostgreSQL schema with anti-clash UNIQUE constraints for production. The store API is identical across dialects, so route code does not change when moving to Prisma/Postgres.

AI Layer

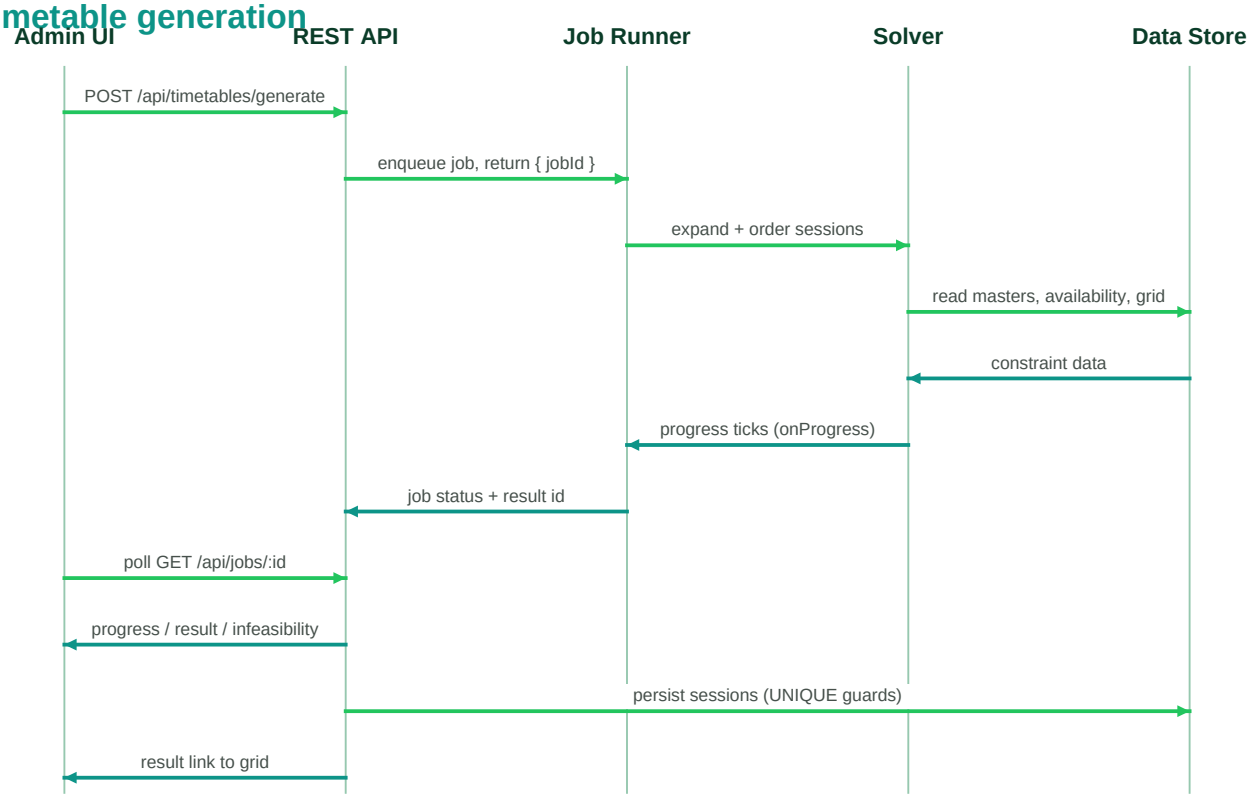
A multi-provider LLM client speaking the OpenAI-compatible chat/completions protocol, plus a deterministic on-device engine that works with no keys at all. Strictly read-only with respect to timetable data.

4. Component Catalog

Component	Responsibility	Key tech
Frontend SPA	Setup wizard, dashboard, generate flow, drag-and-drop grid, Today view, Bulba chat, command palette.	React, Vite, Router, TanStack Query, Zustand, Framer Motion
API server	Mounts routes, validates input, enforces auth/RBAC, returns unified response envelopes.	Node.js, Express, Zod, morgan, cors
Auth service	Login, refresh-token rotation, logout, lockout, password hashing.	jsonwebtoken, bcryptjs, cookie-parser
Master data service	CRUD for teachers, rooms, subjects, batches, assignments, departments.	Express routes + record store
Availability service	Teacher unavailability, requests, HOD approval, read-unification across roles.	Express routes + Zod
Time grid service	Working days, periods per day, period timings, break slots (reserves lunch).	Express routes
Job runner	Queues generation jobs, tracks status/progress, exposes terminal result or report.	In-process async worker
Solver	Backtracking + forward-checking CSP placement; infeasibility detection.	Pure JS
Optimiser	Hill-climbing swaps for soft constraints without breaking hard rules.	Pure JS
Clash auditor	Re-checks teacher/room/batch bookings before publish; failure blocks publish.	Pure JS over store
Export service	CSV and iCal (.ics) generation; PDF returns 501 by design.	Node streams
AI service	Chat, rules parsing, explanations, Q&A; multi-provider with on-device fallback.	llm.js + ai.js
Audit log	Records create/update/delete/move/publish/snapshot/generate/seed/login with who and when.	Record store
Data store	Generic sync record API over SQLite/MySQL; PostgreSQL schema for production.	node:sqlite, mysql2, Prisma

5. Runtime Flows

5.1 Timetable generation



5.2 Authentication and session lifecycle

- **Login.** POST /api/auth/login verifies the bcrypt hash, returns a 15-minute access token and sets a 7-day rotating refresh token in an httpOnly cookie.
- **Refresh.** POST /api/auth/refresh rotates the refresh token and issues a new access token; logout revokes it immediately.
- **Lockout.** Five failed logins within 15 minutes locks the account for 15 minutes and notifies an Admin or HOD.
- **Authorise.** Each request re-checks role and department scope server-side; account deletion is soft (deactivate when sessions exist).

5.3 Manual session editing

- Drag-and-drop or click-to-move calls PATCH /api/sessions/:id with slot_id or day+period (and optionally room_id).
- Every move is validated server-side; blocked moves return a plain-language reason plus suggested valid slots.
- Locked sessions cannot move until unlocked; the same UNIQUE constraints guard the write at the database level.

6. Scheduling Engine

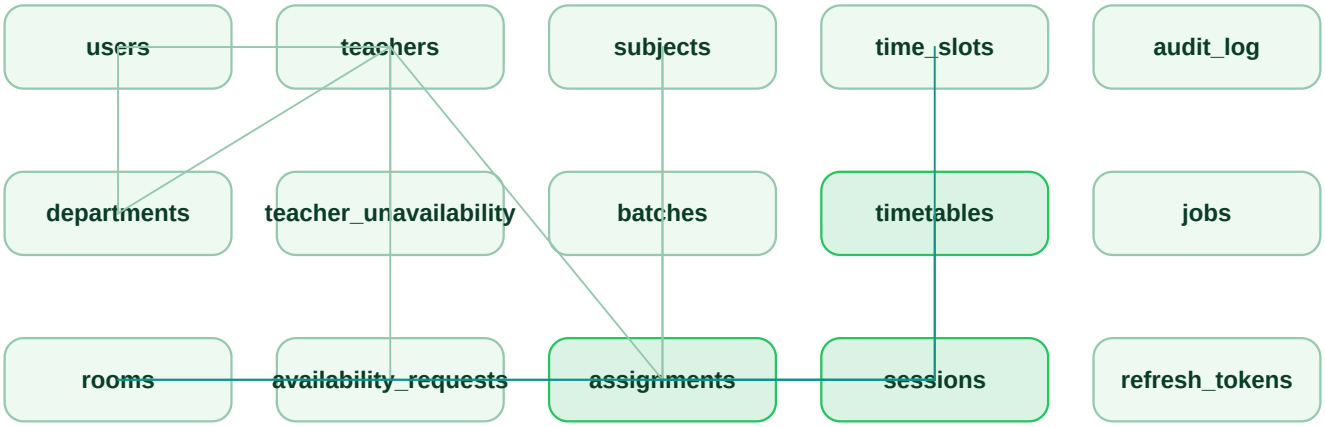
backend/src/solver.js implements constraint-satisfaction timetabling. It is a pure-JS module so it can run inside the API process or be lifted into a worker thread / queue without changing its interface.

1. **Expand**
   Turn every assignment into the individual sessions that must be placed (weekly sessions x periods per session).
2. **Order**
   Sort by difficulty: fewest valid start slots first, with labs/long sessions, tight availability and big batches weighted up.
3. **Search**
   Backtracking with forward-checking (MRV): place the most-constrained session first; placing removes that slot/teacher/room from competitors.
4. **Detect**
   Early infeasibility detection: any session with no valid room or no valid start slot fails fast with a report and the top 3 computed fixes.
5. **Optimise**
   Hill-climbing swaps improve soft constraints (spread subjects, reduce gaps, respect caps) without ever breaking a hard rule.
6. **Flatten**
   Emit one row per period, ready for the database's anti-clash UNIQUE constraints.

Configuration: timeLimitMs (default 25s), maxNodes (default 60,000) and an onProgress callback wired to the job record. Hard constraints are guaranteed: a teacher, room and batch each hold at most one session per slot; availability, room capacity/type and break avoidance are respected; multi-period blocks stay consecutive and never cross a break; and every required weekly session is placed or reported.

7. Data Architecture

The development store (SQLite/MySQL) mirrors the production PostgreSQL tables via a generic synchronous record API, so migration to Prisma/Postgres requires no route changes.



7.1 Anti-clash constraints (on sessions)

These are enforced in the PostgreSQL schema and re-audited in the application before publish:

- UNIQUE (timetable_id, slot_id, teacher_id)
- UNIQUE (timetable_id, slot_id, room_id)
- UNIQUE (timetable_id, slot_id, batch_id)

Multi-period sessions are stored as one row per period, so these constraints cover them too.

7.2 Core collections

departments, users, teachers, teacher_unavailability, availability_requests, rooms, subjects, batches (with parent_batch_id for split labs), assignments, time_slots, timetables, sessions, jobs, audit_log, refresh_tokens.

8. AI Assistant Layer

Bulba is a multilingual, read-only assistant. AI never writes timetable data: the solver and database constraints own correctness, and AI only reads, translates, explains and summarises. It works with no keys at all via an on-device engine and upgrades to a hosted multilingual chatbot when a provider key is added.

8.1 Provider fallback chain

Order	Provider	Config
1. Local VM (primary)	OpenAI-compatible endpoint on your VM; no key needed	LOCAL_AI_BASE_URL / LOCAL_AI_MODEL
2. OpenRouter	Free models behind a compatible base URL	OPENROUTER_API_KEY
3. Google AI Studio	Gemini models	GEMINI_API_KEY
4. NVIDIA NIM	Hosted chat models	NVIDIA_API_KEY

Default order is local -> openrouter -> google -> nvidia, overridable with AI_FALLBACK_ORDER. Only configured providers are in the chain. Any error, timeout or rate-limit rolls to the next provider; each attempt has a 30s timeout. If every provider fails, the endpoint falls back to the on-device engine and marks the reply fallback: true. Responses report provider, tried and fellBack.

8.2 Capabilities and endpoints

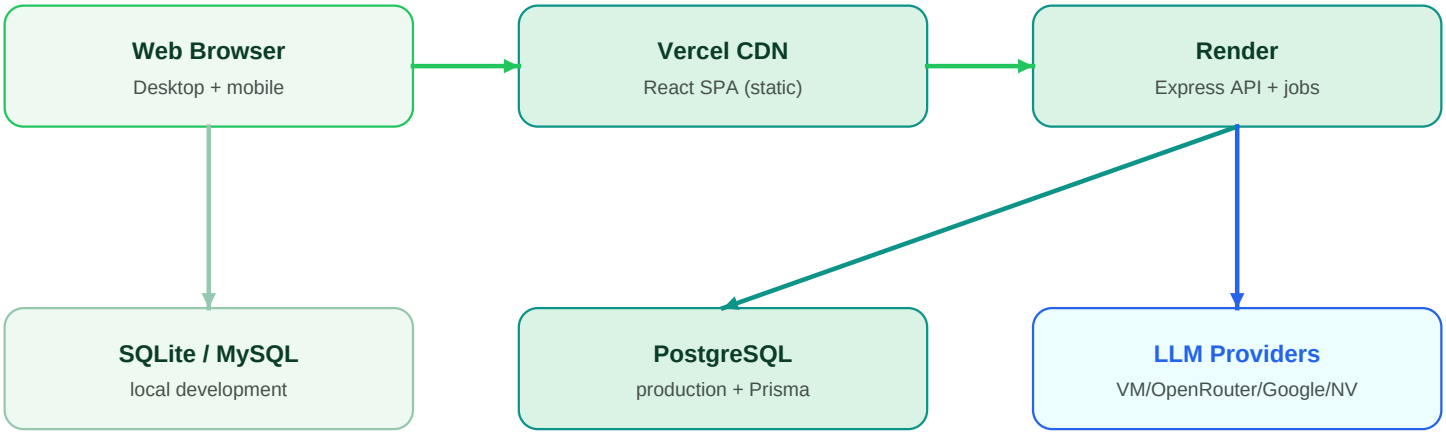
ID	Feature	Endpoint	Status
A0	Multilingual chat (Bulba) with on-device fallback	POST /api/ai/chat	Live
A1	Plain-English rules -> structured availability suggestion	POST /api/ai/rules/parse	Live
A2	Infeasibility explainer (plain words + top fixes)	POST /api/ai/explain	Live
A3	Smart CSV import column mapping and typo flags	POST /api/ai/import/map-columns	Planned
A4	Ask SlotSync natural-language Q&A over the timetable	POST /api/ai/ask	Live
A5	Change notes summary when a timetable changes	POST /api/ai/explain	Live
-	Provider + health for the assistant badge	GET /api/ai/status	Live

8.3 Guardrails

- AI never writes timetable data; every AI output is a suggestion the admin confirms.
- Structured JSON output validated by Zod, with one retry then a friendly fallback to the manual form.
- Only a compact data digest (codes, names, counts) is sent; no student PII.
- Keys are never logged or returned to the client; a 30s timeout protects the UI.

9. Deployment Topology

The API and SPA are independent sites; the LLM endpoints are external or self-hosted.



- **API (Render).** Set FRONTEND_URL to the Vercel origin (comma-separated for several; *.vercel.app previews allowed), a long random JWT_SECRET, and NODE_ENV=production. CORS reflects the allowed origin; the refresh cookie is SameSite=None; Secure.
- **SPA (Vercel).** frontend/.env.production points the build at the API host; vercel.json rewrites all paths to index.html for client-side routing.
- **Database.** Local dev uses node:sqlite or MySQL; production uses PostgreSQL via Prisma (npx prisma migrate deploy). TiDB Cloud is supported as a managed MySQL-compatible option.
- **AI endpoints.** A self-hosted OpenAI-compatible server is primary; hosted keys are optional fallbacks. Keys live only in backend environment variables.

10. Security, Reliability & NFRs

10.1 Security

- Passwords hashed with bcrypt; hashes never returned to clients.
- Short-lived 15-minute access token plus a 7-day rotating refresh token in an httpOnly cookie.
- Role and department checks on every request (Admin global, HOD own department, Teacher/Student scoped).
- Login rate limiting: 5 failed attempts in 15 minutes locks the account for 15 minutes.
- Teachers owning sessions are deactivated, never hard-deleted; users cannot delete themselves or manage admins.
- Every sensitive action is written to the audit log (who, what, when, previous value).

10.2 Reliability and integrity

- Database-level UNIQUE constraints make clashes impossible regardless of code path.
- An automated clash audit re-checks every teacher, room and batch per slot before publish; a failure blocks publishing.
- Multi-step changes run in one transaction, so a failure leaves the timetable unchanged.
- Structured logs and error tracking on the API and solver; nightly database backups.

10.3 Non-functional targets

Area	Target
Scale (v1)	Up to 60 teachers, 40 rooms, 40 batches, 6 days x 8 periods
Generation time	< 30s typical, < 2 min worst case, async with progress
UI performance	Grid interactions at 60 fps; initial load < 2.5s on 4G
Reliability	99.5% uptime during term; nightly DB backups
Correctness	0 hard conflicts in any published timetable (verified by audit)
Browser support	Latest Chrome, Edge, Safari, Firefox; iOS Safari and Android Chrome

10.4 Technology stack summary

Layer	Technology
Frontend	React 18, Vite 5, React Router 6, TanStack Query 5, Zustand 4, Framer Motion 11
Backend	Node.js, Express 4, Zod, jsonwebtoken, bcryptjs, cookie-parser, morgan, cors
Database (dev)	node:sqlite (default) or MySQL (mysql2) via a generic records store
Database (prod)	PostgreSQL via backend/prisma/schema.prisma
Solver	Pure-JS backtracking with forward-checking + hill-climbing (backend/src/solver.js)
AI	Multi-provider LLM client + deterministic on-device engine (backend/src/llm.js, ai.js)